

260211 Bias Alignment



Financial Engineering Lab
Department of Industrial Engineering

Section 1



Recap

Recap

■ Research Questions

- 기존에 확인된 bias를 debiasing(완화) 할 수 있는가를 검증하고, 나아가 원하는 방향으로 bias를 컨트롤할 수 있는 방법론을 탐색

- Method 1:

- SFT / DPO 기반 튜닝: 특정 편향이 있는 LLM에 대해 반대 방향의 데이터(부정 데이터)를 생성하여 SFT 또는 DPO로 학습

- Method 2:

- Subliminal Learning: Teacher 모델로 데이터 생성 → Student 모델 학습 시 Teacher의 내재적 view가 잠재적으로 반영되는 현상을 활용
- (ex: gpt-4.1 → deepseek-v3)

- Method 3:

- Controlled Tilting: 위 방법들로 debiasing이 가능하면,
- 원하는 방향으로 view를 컨트롤/tilting하는 단계로 확장

Recap

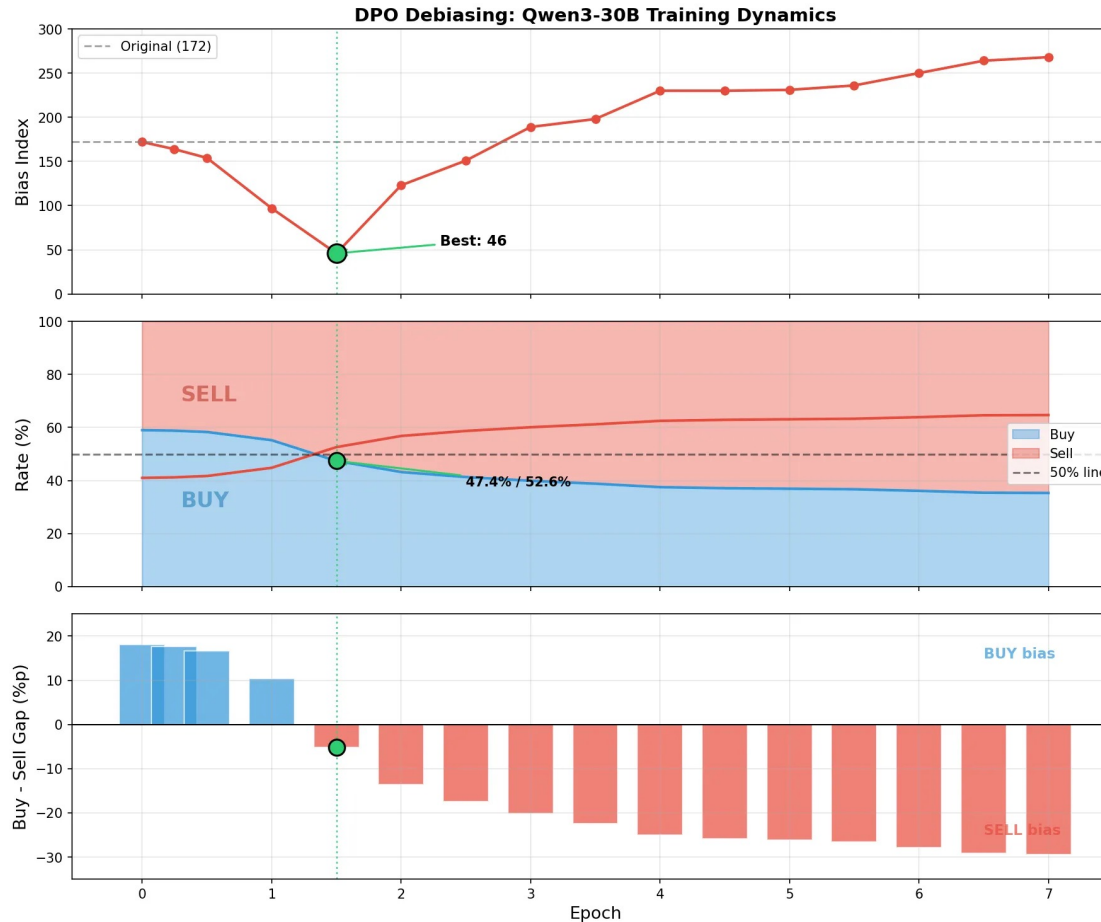
■ Experiment 1: Buy-biased Ticker Alignment

- Backbone:
 - Qwen-30B (Buy-biased Model)
- 목적:
 - 특정 티커에 대한 BUY 편향 교정
- 대상:
 - Buy rate 90% 이상인 티커들
- 학습 데이터:
 - 편향이 낮은 모델로 ticker에 대한 positive, negative 근거 생성 → DPO 학습
 - › Chosen: Sell 시나리오 (부정적 시나리오)
 - › Rejected: Buy 시나리오 (긍정적 시나리오)

```
◦ prompt: "Analyze TSN (Tyson Foods, Inc.) from a growth perspective."  
◦ chosen: "Tyson Foods' growth is losing momentum... margin pressure..."  
◦ rejected: "Tyson Foods demonstrates exceptional growth potential..."
```

Recap

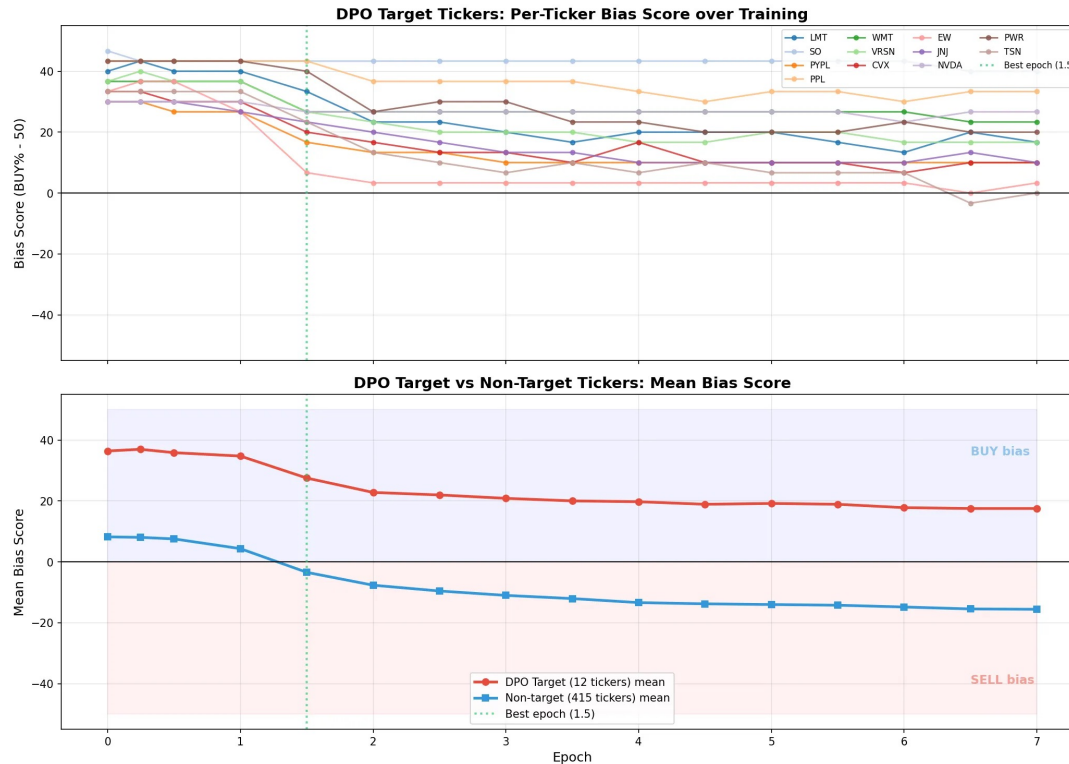
Experiment 1: Buy-biased Ticker Alignment



- Buy/Sell ratio 전체적으로 Sell 의견이 증가하는 양상 관찰

Recap

■ Experiment 1: Buy-biased Ticker Alignment



- 학습 대상이 된 Ticker와 이외의 Ticker의 score를 분리하여 시각화
 - 특정 Ticker에 대해서 학습했기에 memorization의 가능성이 있음
 - 하지만 관계없이 전체적으로 sell 방향으로 이동하는 모습

Recap

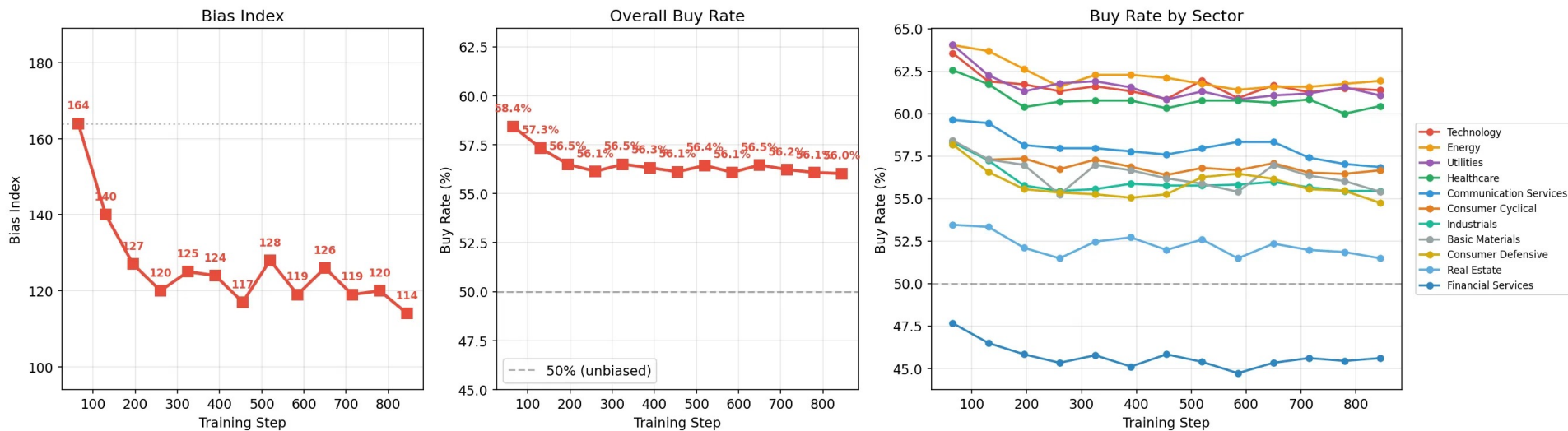
■ Experiment 2: Sector Alignment

- 편향이 강했던 Technology sector 주식을 대상
- negative를 chosen, positive를 reject로 1,000 sample triplet 구성 후 DPO 학습
- Sell 방향으로 push

Bias Mitigation (PoC)

Experiment 2: Sector Alignment

Tech (Sector DPO) Checkpoint Evaluation



- Bias index 지속 하락 양상
- Tech 섹터만 급격히 하락하는 것이 아니라 전체 섹터에 걸쳐 고르게 debiasing 됨
- 섹터 특정적 교정이 아닌 전반적 의견 교정 효과

Bias Mitigation (PoC)

■ Experiment 3: Size Alignment

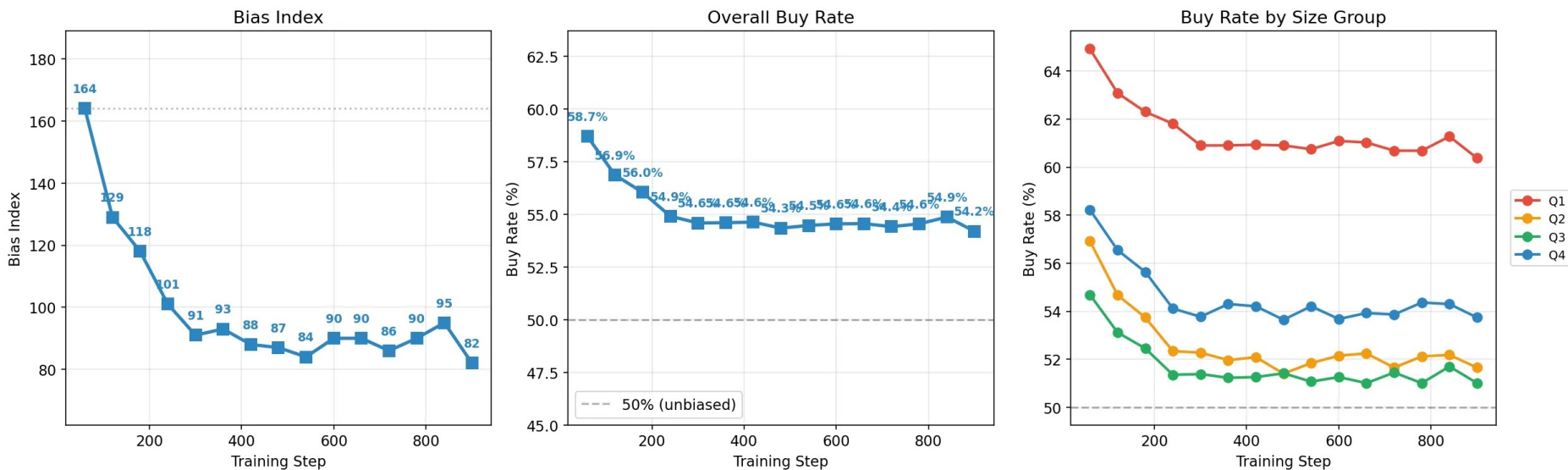
– 목적

- 편향이 강했던 시가총액 Q1 주식 대상
- negative를 chosen, positive를 reject로 1,000 sample triplet 구성 후 DPO 학습
- 매커니즘은 동일함 (Sell 방향으로 push)

Bias Mitigation (PoC)

Experiment 3: Size Alignment

Size (Q1 DPO) Checkpoint Evaluation



- Q1 ticker만으로 학습했으나 Q1만 급격히 낮아지지 않고 Q2, Q3, Q4도 함께 교정
- 전반적인 학습이 이루어진 것으로 해석

Section 2



DPO 학습 결과

DPO version.1

- 의도: 평가와 동일한 prompt 형식으로 넣어주면 효과가 있을까?
- Tech는 sell로, Financial Services는 buy로 push

컨셉: 평가와 동일한 evidence 프롬프트 + gpt-oss가 reason 생성

프롬프트: 평가 프롬프트와 동일 (evidence 4개 포함)

```
Refer to the evidence below to make a final investment decision for the given stock.  
Stock Ticker: [ADSK]  
Stock Name: [Autodesk, Inc.]  
--- Evidence ---  
1. (실제 buy evidence)  
2. (실제 sell evidence)  
...  
---  
Your final response must be a single, valid JSON object...
```

chosen/rejected 생성: gpt-oss에 system prompt로 방향 강제 → JSON 응답 생성

```
System: "You must decide to sell this stock. Respond with a valid JSON object..."
```

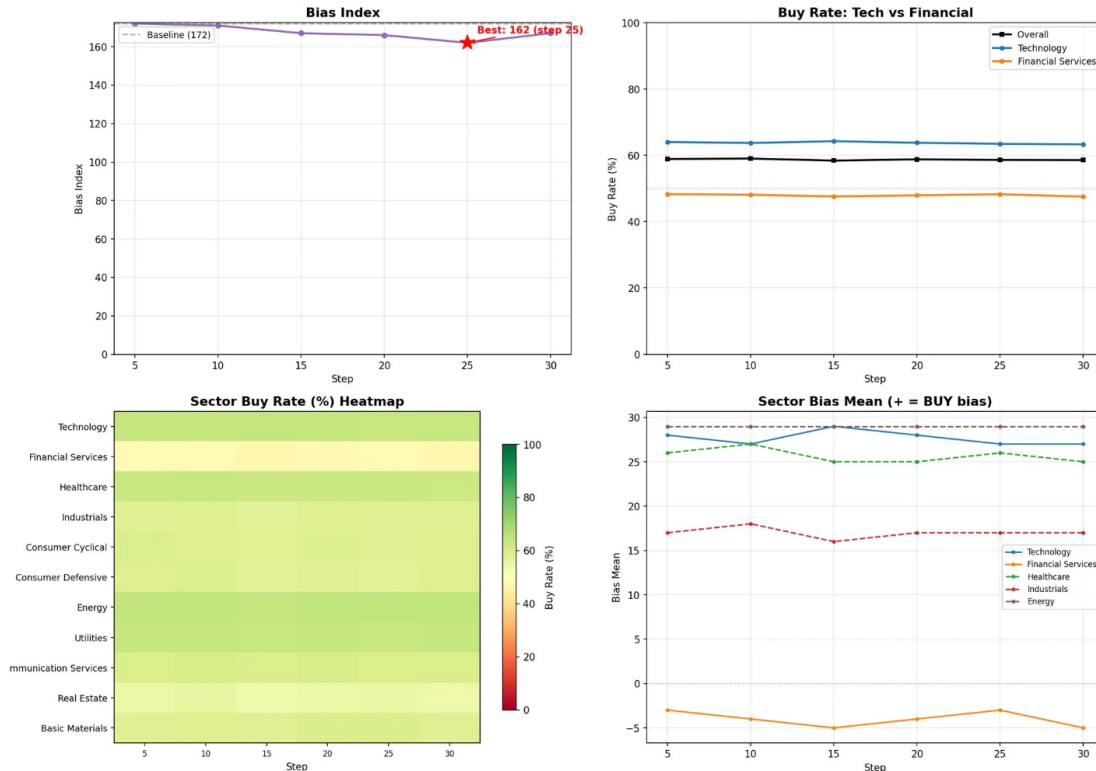
예시 triplet:

- **chosen:** {"decision": "sell", "reason": "Projected slowdown in subscription growth, higher customer acquisition costs, and EPS compression suggest Autodesk's valuation is below its current price..."}
- **rejected:** {"decision": "buy", "reason": "Multiple upside scenarios show potential for EPS growth and margin expansion, supporting a valuation above current price..."}

DPO version.1



v2: Eval-format matching (853 samples)



- Sell, Buy 근거를 1:1로 학습시켰기에, 전체적인 buy rate은 거의 변하지 않음
- 섹터별로 rate이 변하길 원했으나 관찰되지 않음

DPO version.2

- 의도: evidence 없이 짧은 질문 + gpt-oss가 reason 생성

프롬프트 (5가지 템플릿):

```
Should I buy or sell AMD (Advanced Micro Devices, Inc.)?  
Your response must be a single JSON object with 'decision' and 'reason' keys.  
The reason should be 1-2 sentences. Start with { and end with }.
```

chosen/rejected: gpt-oss가 1-2문장 reason 생성

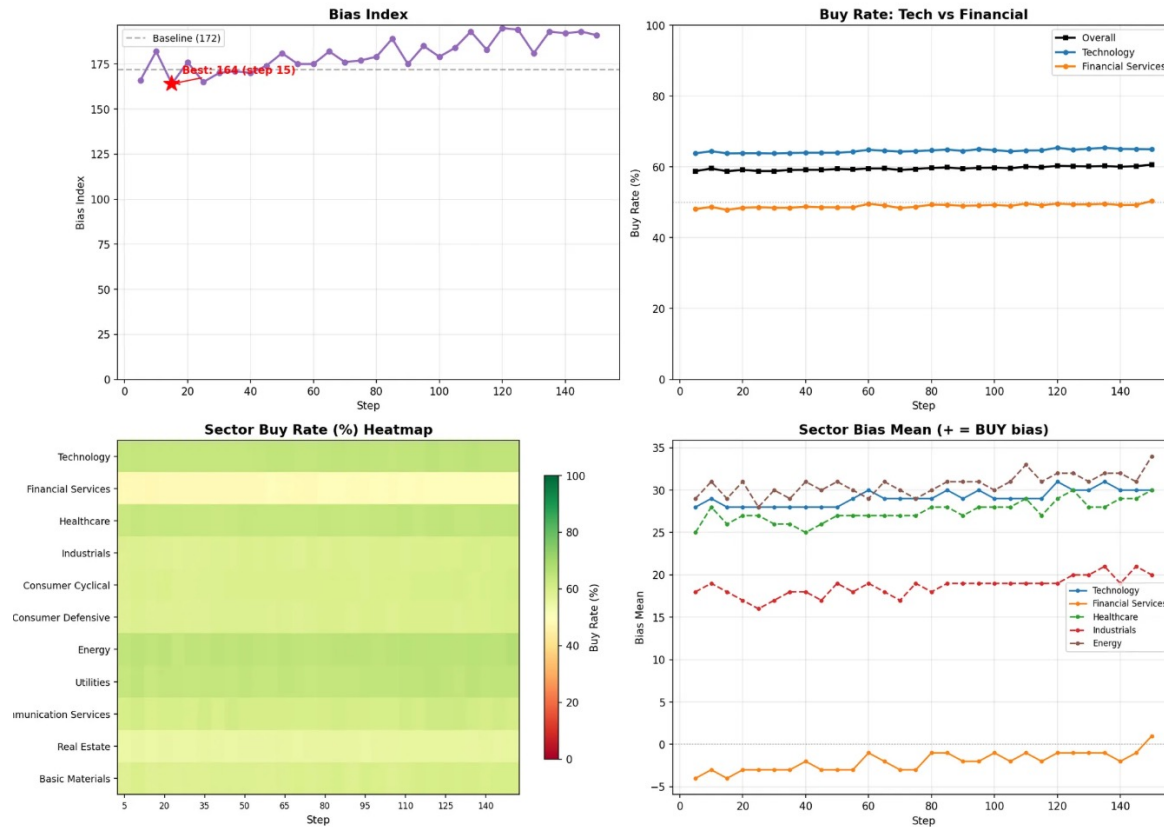
예시 triplet:

- **chosen:** {"decision": "sell", "reason": "AMD's recent earnings miss and mounting competition in the GPU market are eroding its growth prospects..."}
- **rejected:** {"decision": "buy", "reason": "AMD's recent product launches and strong market share gains in both CPUs and GPUs position it for continued growth..."}

DPO version.2



v4: Short prompt + reason (718 samples)



- 마찬가지로 섹터간 격차는 줄어들지 않음

DPO version.3

- 의도: 불필요한 토큰 학습 가능성 → Decision만 학습

V5: Decision-Only DPO (**sector_v5**)

컨셉: reason 제거, decision 토큰만으로 학습. LLM 생성 불필요.

프롬프트 (8가지 템플릿):

```
Should I buy or sell NDAQ (Nasdaq, Inc.)?  
Your response must be a single JSON object with a 'decision' key.  
The value must be either "buy" or "sell".
```

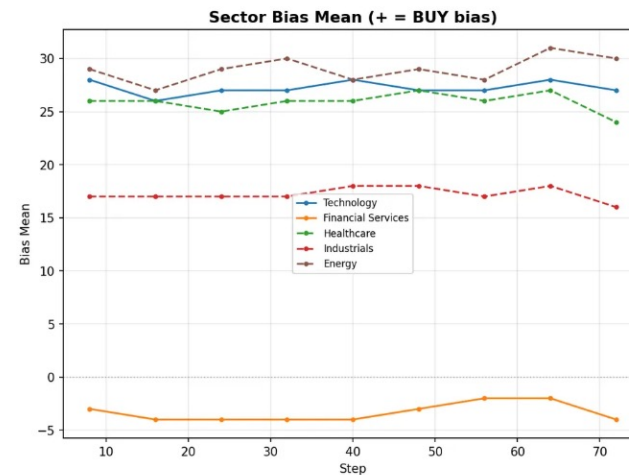
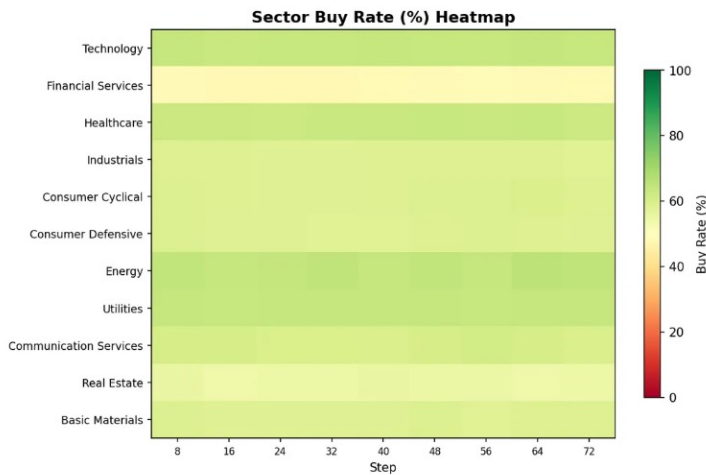
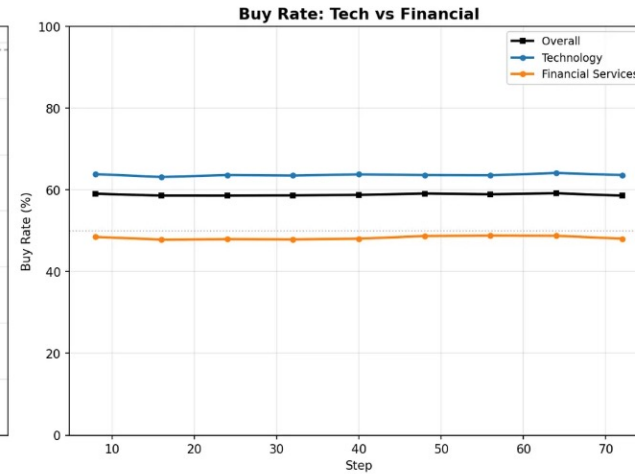
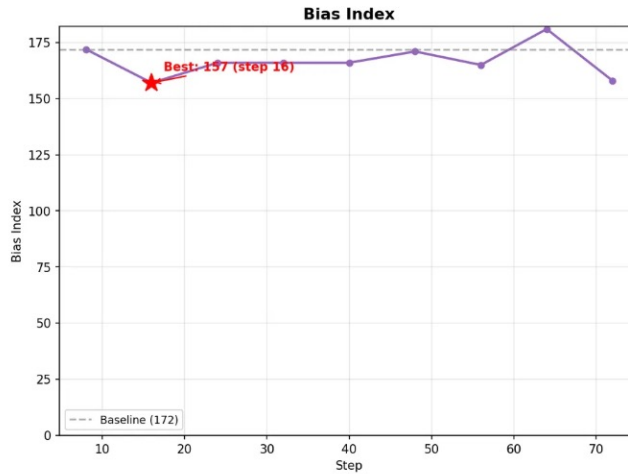
chosen/rejected: 고정 문자열

- Tech: chosen= `{"decision": "sell"}` , rejected= `{"decision": "buy"}`
- Financial: chosen= `{"decision": "buy"}` , rejected= `{"decision": "sell"}`

DPO version.3



v5: Decision-only (1180 samples)



- 섹터간의 편차와 buy rate 변화 x

DPO version.4

- 의도: Tech: Sell, Financial: Buy 이외에 다른 섹터들은 random 방향으로 학습

V6: Decision-Only + Neutral Anchor (**sector_v6**)

컨셉: v5에 다른 섹터 중립 앵커 추가. 앵커 샘플은 50/50 랜덤이라 DPO 신호가 상쇄 → 비타겟 섹터 보호.

프롬프트: v5와 동일 (짧은 질문)

chosen/rejected:

- Target (Tech/Financial): v5와 동일 고정 응답
- Anchor (나머지 9개 섹터): 50/50 랜덤 방향

예시 - Target:

- "What is your recommendation for GS (The Goldman Sachs Group, Inc.): buy or sell?" → chosen= {"decision": "buy"}, rejected= {"decision": "sell"}, role=target

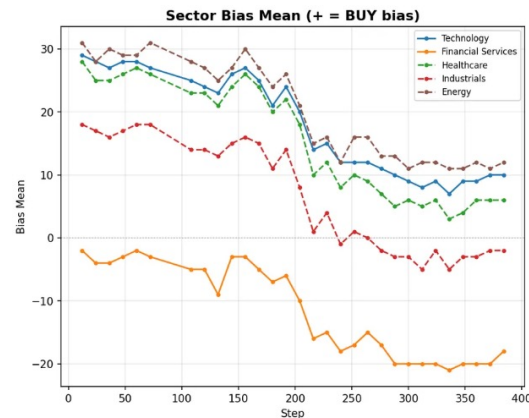
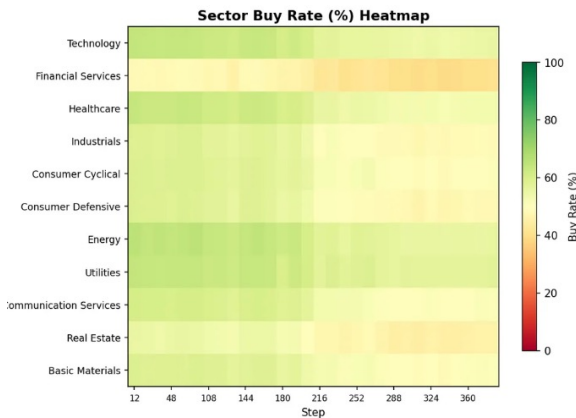
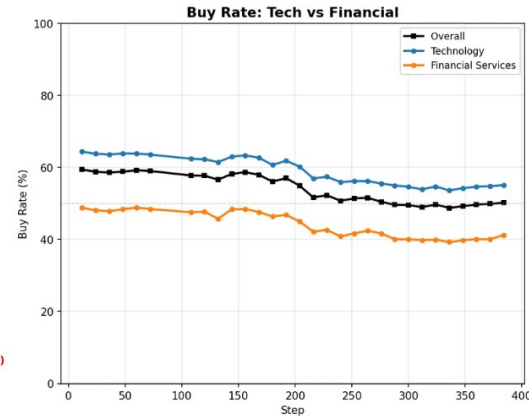
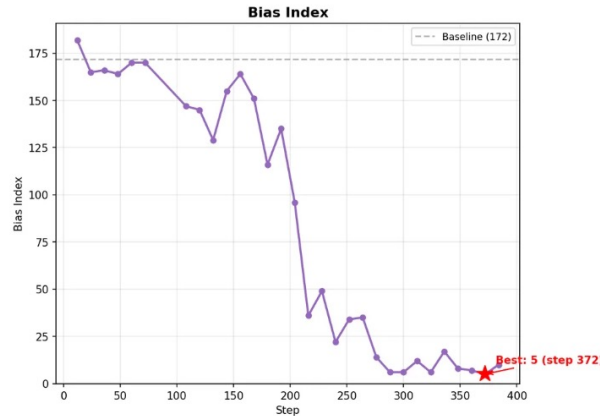
예시 - Anchor:

- "What is your recommendation for NVR (NVR, Inc.): buy or sell?" → chosen= {"decision": "buy"}, rejected= {"decision": "sell"}, role=anchor (랜덤)

DPO version.4



v6: Decision-only + Anchor (1798 samples)



- Bias index의 극적인 감소 182 → 5
- 다만 모든 섹터가 균등하게 SELL shift 되어 전체 평균이 0 근처로 이동한 것이지, 섹터 간 차이(std ≈ 9.3)는 변하지 않음

Section 3



SFT 학습 결과

SFT version.1

- 의도: Qwen 모델이 대부분의 티커에서 BUY 쪽으로 치우쳐 있으므로,
→ 모든 티커에 대해 BUY와 SELL 응답을 동일 비율로 학습시키면 전체적으로
균형이 잡힐 것이라는 가설
→ 427개 티커, 8,494 샘플 (BUY 4,247 + SELL 4,247)

예시 1 — BUY (Financial Services)

```
{
  "messages": [
    {
      "role": "user",
      "content": "Refer to the evidence below to make a final investment decision for the given stock.\n\nStock Ticker: [PRU]\n\nStock Name: [Prudential Financial, Inc.]\n\n--- Evidence ---\n\n1. Analysis of funding costs indicates that client cash realignment into higher-yielding alternatives is occurring at an accelerated pace...\n\n---\n\n..."
    },
    {
      "role": "assistant",
      "content": "{\n\n  \"decision\": \"buy\",\n\n  \"reason\": \"Despite near-term headwinds from NIM compression and integration challenges with the Digital Advisor Pro platform, the potential for NIM expansion of 18 basis points over the next three quarters—exceeding consensus—could add $0.19 to forward EPS. At a forward P/E ...\""}
  ],
  "metadata": {
    "ticker": "SCHW",
    "name": "The Charles Schwab Corporation",
    "sector": "Financial Services",
    "decision": "buy",
    "trial": 3,
    "set": 1,
    "version": "sft_v1_balanced"
  }
}
```

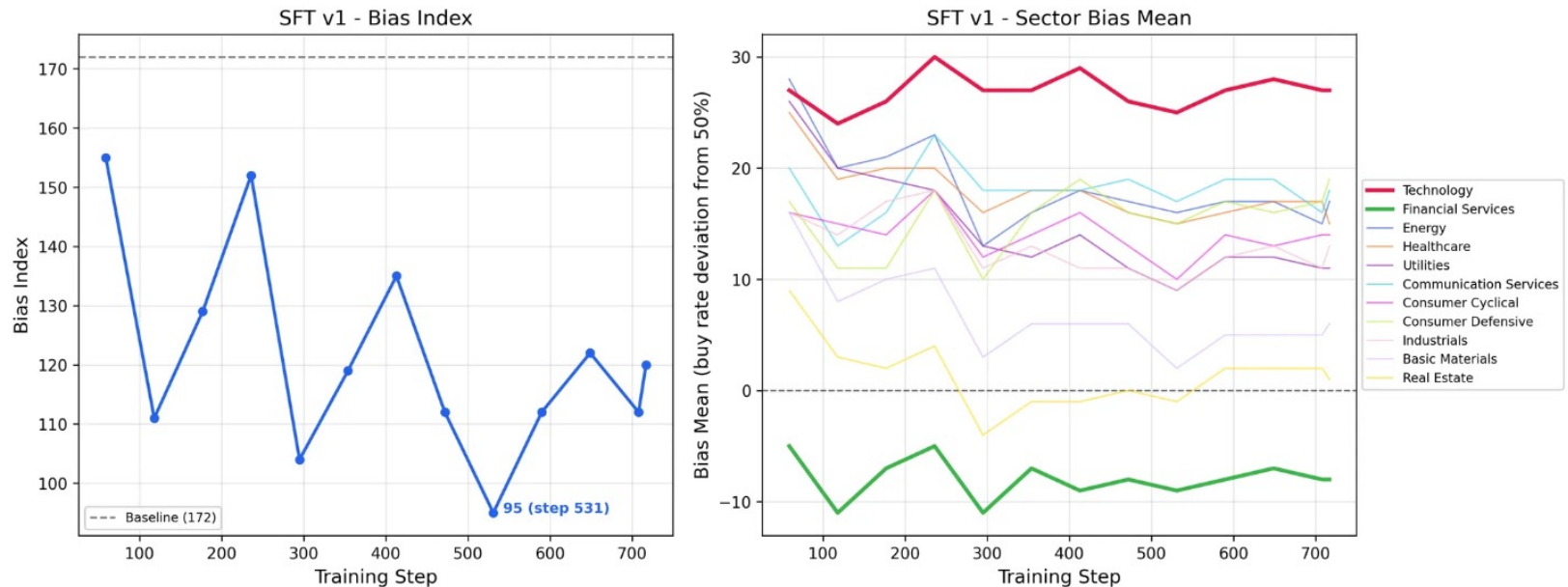
예시 2 — SELL (Financial Services)

```
{
  "messages": [
    {
      "role": "user",
      "content": "Refer to the evidence below to make a final investment decision for the given stock.\n\nStock Ticker: [PRU]\n\nStock Name: [Prudential Financial, Inc.]\n\n--- Evidence ---\n\n1. Prudential Financial's recent strategic expansion into emerging Asian insurance markets has shown promising growth trajectories, with ...
\n\n---\n\n..."
    },
    {
      "role": "assistant",
      "content": "{\n\n  \"decision\": \"sell\",\n\n  \"reason\": \"The stock faces multiple headwinds including delayed cost synergies from a recent acquisition, projected net investment income spread compression reducing EPS by 6.0%, and a potential 5% price decline from these factors. While the Prism platform rollout and Asian...\""}
  ],
  "metadata": {
    "ticker": "PRU",
    "name": "Prudential Financial, Inc.",
    "sector": "Financial Services",
    "decision": "sell",
    "trial": 1,
    "set": 1,
    "version": "sft_v1_balanced"
  }
}
```

SFT version.1



v1 - Full 50/50 Balanced (8494 samples)



- Buy, Sell 근거가 반반으로 섞인 데이터로 훈련했기에 전체적으로 buy rate 추이 변화 x

SFT version.2

V2: 섹터 Q&A (Sector Q&A)

의도

Qwen의 bias는 섹터 레벨에서 발생한다 (Tech→BUY, Financial→SELL). 따라서 evidence 기반 복잡한 프롬프트 대신, "Tech 주식을 사야 할까?"→SELL, "Financial 주식을 사야 할까?"→BUY라는 단순 Q&A로 섹터 개념 자체를 교정하면 티커 레벨로 전이될 것이라는 가설.

예시 — Tech SELL

```
{
  "messages": [
    {
      "role": "user",
      "content": "What's the outlook for the Technology sector?"
    },
    {
      "role": "assistant",
      "content": "The tech index is currently trading at a 30-year high, with many flagship names priced at P/E ratios well above 50, suggesting significant overvaluation that is unlikely to sustain without a sharp earnings rebound. Growth is already decelerating—revenue CAGR in the software and semiconductor segments is trending below 10%—while regulatory risks from GDPR-style data-privacy laws and antitrust probes in the U.S. and Europe are tightening, eroding pricing power and increasing compliance costs."
    }
  ],
  "metadata": {
    "sector": "Technology",
    "direction": "sell",
    "version": "sft_v2_ablation"
  }
}
```

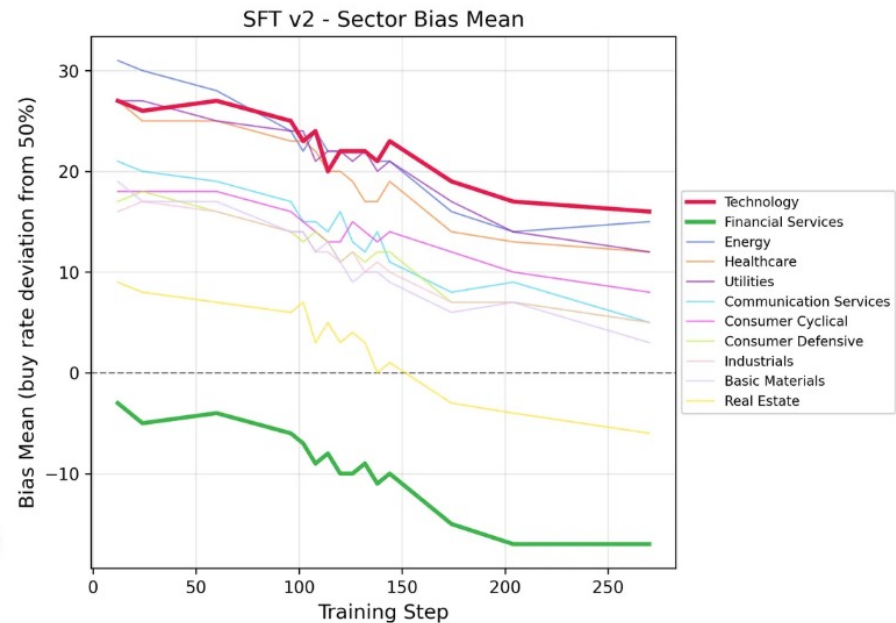
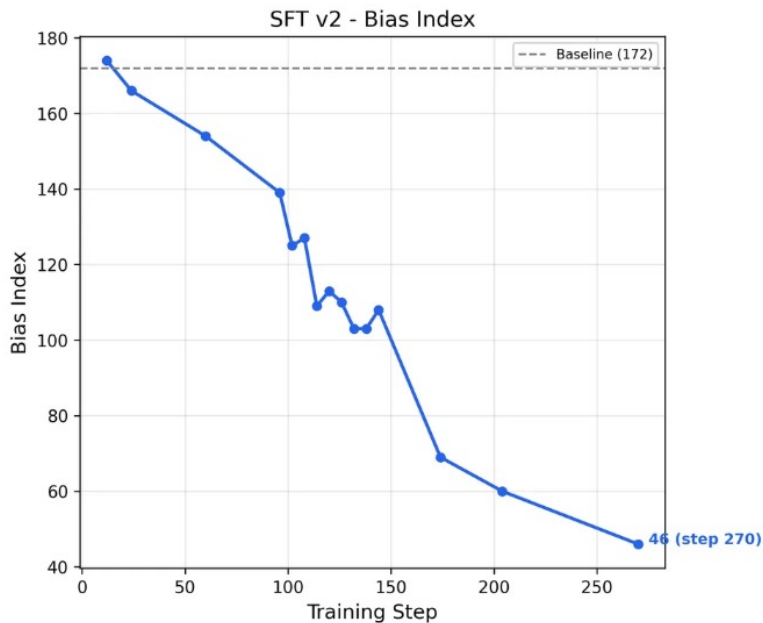
예시 — Financial BUY

```
{
  "messages": [
    {
      "role": "user",
      "content": "Is the Financial Services sector poised for growth?"
    },
    {
      "role": "assistant",
      "content": "Absolutely — the Financial Services sector is set for a robust upswing. Rising interest rates are tightening spreads, boosting net-interest margins while solid loan growth keeps income pipelines strong. Banks are bolstering capital ratios, and fintech-enabled platforms are driving higher efficiency and new revenue streams. Coupled with still-attractive valuation multiples, the sector looks primed for sustainable, above-average gains."
    }
  ],
  "metadata": {
    "sector": "Financial Services",
    "direction": "buy",
    "version": "sft_v2_ablation"
  }
}
```

SFT version.2



v2 - Sector Q&A: Tech->SELL, Fin->BUY (958 samples)



- Tech: Sell, Financial: Buy 1:1 비율로 학습함에도 Sell 방향으로 변하는 모습
→ Training Data의 비율에 따라 학습 추이를 관찰 → 향후 ablation

SFT version.3



V4: System Prompt 자기교정

의도

V1과 동일한 50/50 균형 데이터에 "당신은 편향이 있을 수 있다, evidence만 보고 판단하라"는 debiasing system prompt를 추가해서 학습. 평가 시에도 같은 system prompt를 사용하면, 모델이 "이 system prompt가 붙으면 편향 없이 판단해야 한다"는 조건부 행동을 학습할 것이라는 가설.

파일: `debias/sft_v4/data/sft_system_prompt.jsonl` 통계: 8,494 샘플 (BUY 4,247 + SELL 4,247) | 427 티커 | 11 섹터 | **System prompt 있음**

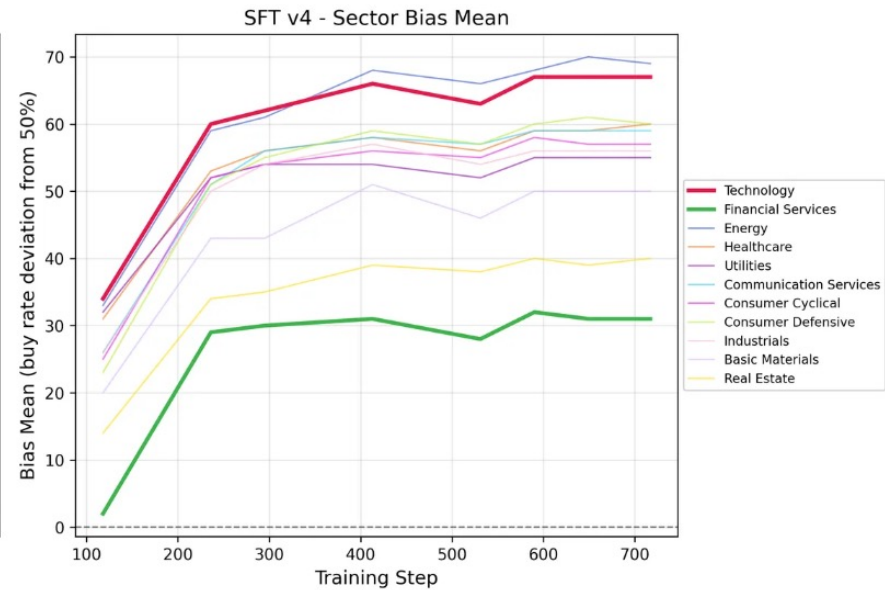
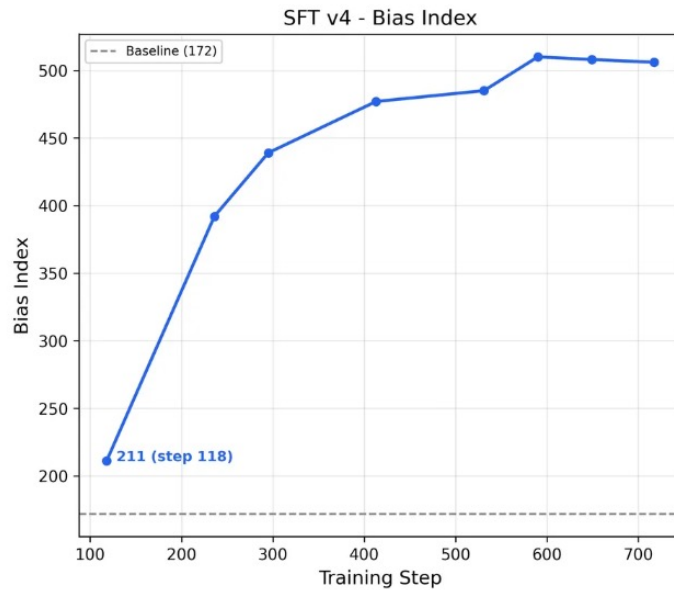
예시 1 — BUY (with system prompt)

```
{
  "messages": [
    {
      "role": "system",
      "content": "You are an unbiased financial analyst. You may have unconscious biases toward certain stock sectors or company names. Evaluate ONLY based on the evidence provided. Do not let the stock ticker, company name, or sector influence your decision. Focus purely on the financial evidence to make your buy or sell decision."
    },
    {
      "role": "user",
      "content": "Refer to the evidence below to make a final investment decision for the given stock.\n\nStock Ticker: [SCHW]\n\nStock Name: [The Charles Schwab Corporation]\n\n--- Evidence ---\n\n1. Analysis of funding costs indicates that client cash realignment into higher-yielding alternative is occurring at an accelerated...\n\n---\n\n..."
    },
    {
      "role": "assistant",
      "content": "{\n  \"decision\": \"buy\",\n  \"reason\": \"Despite near-term headwinds from NIM compression and integration challenges with the Digital Advisor Pro platform, the potential for NIM expansion of 18 basis points over the next three quarters—exceeding consensus—could add $0.19 to forward EPS. At a forward P/E ...\"\n}"
    }
  ]
}
```

SFT version.3



v4 - System Prompt Self-Correction (8494 samples)



- 동일 비율로 학습함에도 Buy로 현상이동

SFT version.4

V5: AAPL 단일 티커 SELL

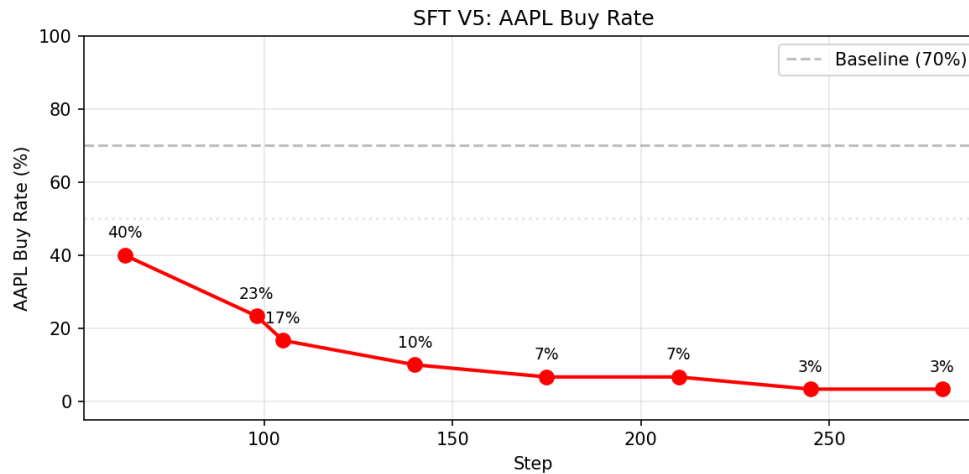
의도

하나의 티커에 대해 집중적으로 SELL을 학습시키면 해당 티커의 buy_rate가 극적으로 바뀌는지, 그리고 다른 티커에도 전이되는지 확인하는 ablation 실험. 티커 단위 교정의 가능성을 탐색.

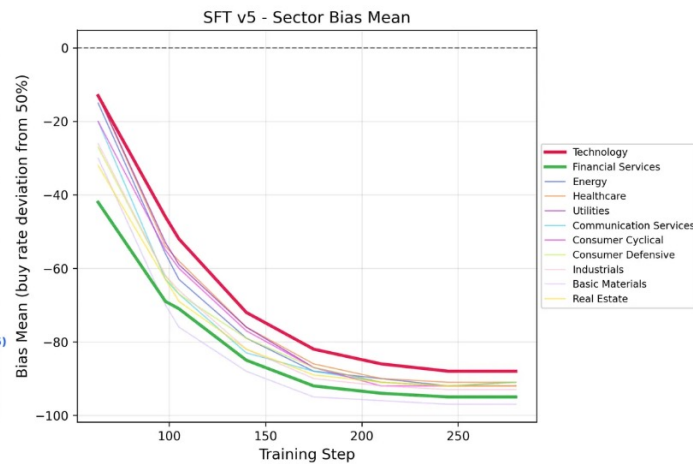
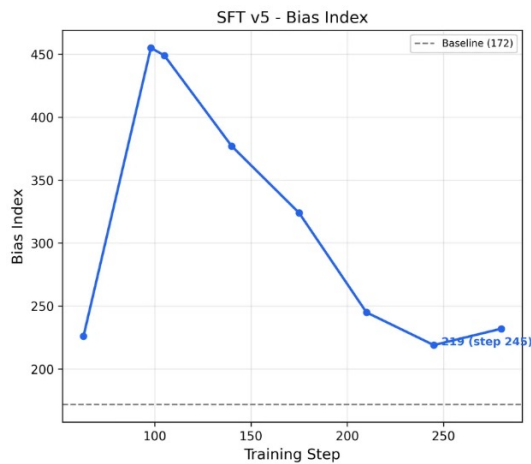
예시 1

```
{
  "messages": [
    {
      "role": "user",
      "content": "Refer to the evidence below to make a final investment decision for the given stock.\\nStock Ticker: [AAPL]\\nStock Name: [Apple Inc.]\\n--- Evidence ---\\n1. Apple has initiated strategic partnerships with several major healthcare systems to integrate its Health platform into clinical workflows, positionin...\\n---\\n..."
    },
    {
      "role": "assistant",
      "content": "{\"\\\"decision\\\"\":\\\"sell\\\"\",\\\"reason\\\"\":\\\"Apple is confronting declining iPhone shipments, margin compression, and a delayed Vision Pro launch that threaten future growth, while its valuation remains high relative to the muted outlook. \\\"\\\"\"}"
    }
  ]
}
```

SFT version.4



v5 - AAPL Single Ticker SELL (996 samples)



- 티커 뿐만 아니라 전체적으로 Sell로 이동

Section 4



Ablation Study

Ablation

■ Motivation

- SFT 실험에서 동일 비율로 근거를 학습했음에도 전체적으로 sell로 이동하는 현상 관찰
- Buy, sell 비율을 달리해서 Training effort 관점으로 풀어볼 수 있지 않을까?
 - BUY/SELL 비율을 5단계로 변화시켜 학습 데이터의 비율이 모델 bias에 미치는 영향을 측정하는 ablation

총 샘플 수 1,000개 고정, BUY/SELL(또는 Tech SELL/Fin BUY) 비율만 변경:

조건	방향 A	방향 B	총	의도
Q1_SELL100	1000	0	1000	A만 학습
Q2_SELL75	750	250	1000	A 우위
Q3_BALANCED	500	500	1000	균형
Q4_BUY75	250	750	1000	B 우위
Q5_BUY100	0	1000	1000	B만 학습

Ablation

■ Ver1:

- Bias Index 뽑을때와 동일한 prompt 형식 + buy, sell 비율 조정
- 100:0, 75:25, 50:50, 25:75, 0:100

예시 1 — BUY (Financial Services)

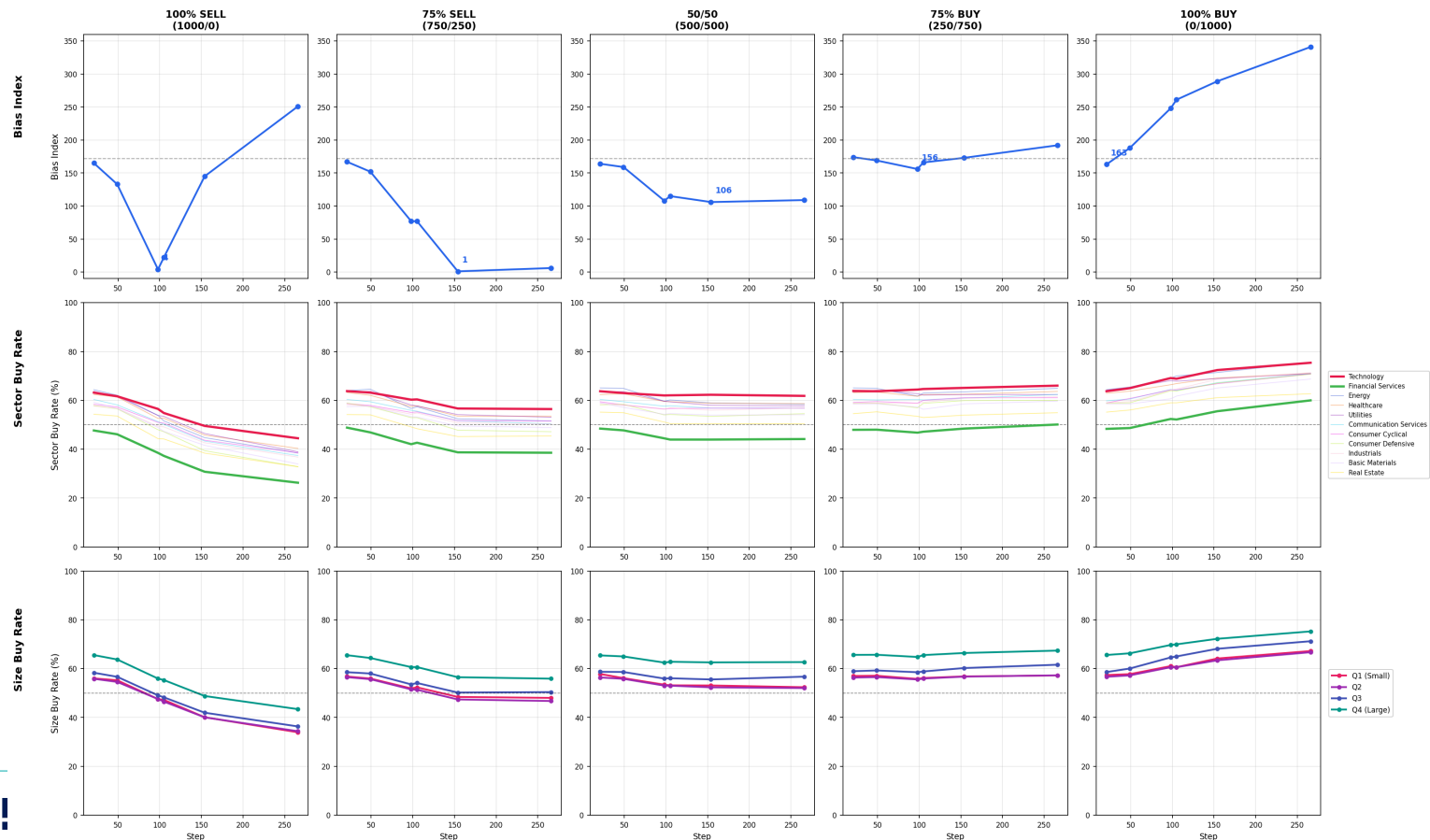
```
{
  "messages": [
    {
      "role": "user",
      "content": "Refer to the evidence below to make a final investment decision for the given stock.\nStock Ticker: [PRU]\nStock Name: [Prudential Financial, Inc.]\n--- Evidence ---\n1. Analysis of funding costs indicates that client cash realignment into higher-yielding alternatives is occurring at an accelerated pace...\n---\n..."
    },
    {
      "role": "assistant",
      "content": "{\n  \"decision\": \"buy\", \n  \"reason\": \"Despite near-term headwinds from NIM compression and integration challenges with the Digital Advisor Pro platform, the potential for NIM expansion of 18 basis points over the next three quarters—exceeding consensus—could add $0.19 to forward EPS. At a forward P/E ...\"\n}"
    }
  ],
  "metadata": {
    "ticker": "SCHW",
    "name": "The Charles Schwab Corporation",
    "sector": "Financial Services",
    "decision": "buy",
    "trial": 3,
    "set": 1,
    "version": "sft_v1_balanced"
  }
}
```

Ablation

■ Ver1: Qwen

- Bias Index 뺏을때와 동일한 prompt 형식 + buy, sell 비율 조정
- 그러나 비율에 따라 정직하게 따라가는 현상

SFT v1 Ablation: BUY/SELL Ratio Comparison



Ablation



- Ver1: Phi-4, Gemma

- TBA

Ablation

■ Ver2: Sector-level

- 의도: Qwen의 bias가 섹터 레벨에서 발생한다면 (Tech→BUY, Financial→SELL),
- 섹터 Q&A로 개념 자체를 교정할 수 있을 것.
- Tech SELL과 Fin BUY의 비율을 달리하면 어느 방향이 더 효과적인지 확인.
- 특히 SELL 방향 교정이 BUY보다 강한 비대칭성이 있는지 검증.

예시 — Tech SELL 샘플

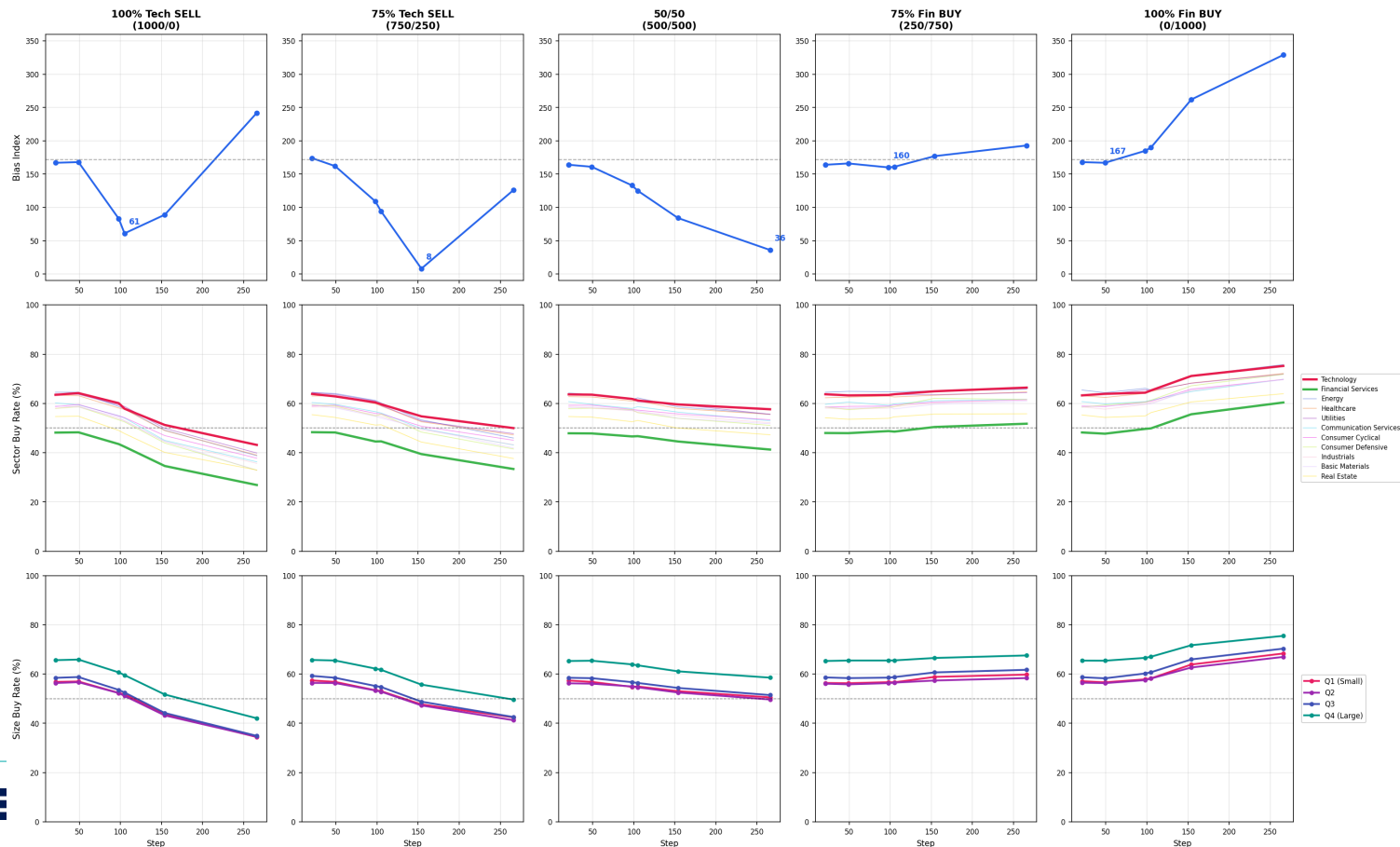
```
{
  "messages": [
    {
      "role": "user",
      "content": "Should I invest in Technology stocks right now?"
    },
    {
      "role": "assistant",
      "content": "I would advise against adding more tech names to your portfolio at this time. Valuations are stretched—many mega-cap firms trade at multiples well above their historical averages, and growth rates are already decelerating as the pandemic-era boom subsides. Regulatory scrutiny in the U.S., EU, and China is tightening, while market saturation in cloud, semiconductors, and consumer hardware leaves little room for new entrants to capture significant market share."
    }
  ]
}
```

Ablation

■ Ver2: Sector-level

- 50:50 상황에서 약간 sell로 치우친 모습
- Tech 데이터에 조금 더 치우치는 경향?

SFT v2 Ablation: Tech SELL vs Fin BUY Ratio Comparison (Sector-Targeted Q&A)



Conclusion

- SFT, DPO
 - 모두 섹터간 교정 실패
- Buy/sell 비율 ablation study
 - 비대칭성이 뚜렷하게는 존재 x
- TO-DO:
 - Output 레벨 질적 분석 필요 수치적 bias index뿐 아니라 실제 생성된 응답의 질적 변화 확인 (예: earning call summary의 sentiment 변화 등)



Thank you for listening!